Teorie pole
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Teorie pole

Princip extremální akce

- prostor historií $\mathcal{H}$
  - "trajektorie"
- funkcionál akce $S: \mathcal{H} \to \mathbb{R}$
  - na konečném čase dává konečnou hodnotu
- klasické řešení (nekvantové) hledá ty správné historie
  - → jsou dány podmínkou
$$\delta S[h] = 0 \quad \text{při fixovaných o.p.}$$
- typicky, ale ne nutně se historie volí trajektorií $x^a(t)$

Lagrangeovský formalismus

- skutečná trajektorie

$$S[x(t)] = \int_{t_2}^{t_1} L(x^a(t), \dot{x}^a(t), t)\, dt$$

může

argumenty jsou obecně nezávislé

$L(x^a, v^a)$

- Lagrangián – x^a ... polohy
  v^a ... rychlosti

⇒ pohybové rovnice

$$\frac{d}{dt}\left(\frac{\partial L}{\partial v}(x,\dot{x})\right) - \frac{\partial L}{\partial x}(x,\dot{x}) = 0$$

Hamiltonovský formalismus

- fázový prostor $[x^a, p_a]$
- pozorovatelné $F(x^a, p_a)$
- polohy x^a nemusí mít vektorovou strukturu
- vždy platí symplektická struktura daná Poissonovými závorkami:

$$\{F,G\} = \sum_a \left(\frac{\partial F}{\partial x^a}\frac{\partial G}{\partial p_a} - \frac{\partial F}{\partial p_a}\frac{\partial G}{\partial x^a}\right)$$

- speciální pozorovatelná – Hamiltonián $H(x^a, p_a)$, platí:

$$\dot{F} = \{F, H\}$$
$$\dot{x}^a \to \dot{x}^a = \{x^a, H\} = \frac{\partial H}{\partial p_a}$$
$$\dot{p}_a \to \dot{p}_a = \{p_a, H\} = -\frac{\partial H}{\partial x^a}$$

- souvislost s Lagrangem:
  $x, v \leftrightarrow x, p$ $\quad L \leftrightarrow H$
  pro nedegenerované problémy
  - Legendreova transformace

$$p = p(x,v) \qquad p_a(x,v) = \frac{\partial L}{\partial v^a}(x,v)$$
$$v = v(x,p) =$$

Chceme: $v(x, p(x,v)) = v \qquad p(x, v(x,p)) = p$

$$\Rightarrow \quad H(x,p) = \sum_a p_a v^a(x,p) - L(x, v(x,p))$$

lze obrátit
$$L(x,v) = \sum_a p_a(x,v)\, v^a - H(x, p(x,v))$$

Polní teorie – funkcionální prostory

- M prostor funkcí na M : $\mathcal{F}M$
- $\mathcal{F}M$ vektorový prostor + topologické vlastnosti
- $\varphi \in \mathcal{F}M \qquad \varphi(x) \in \mathbb{R} \qquad x \in M$
  → $\varphi^x := \varphi(x)$ je komponenta vektoru
- $\omega \in (\mathcal{F}M)^* \qquad \omega_x$

$$\langle \omega, \varphi \rangle = \sum_x \omega_x \varphi^x = \int \boxed{\omega_x} \boxed{\varphi^x} \boxed{dV}$$

ve skutečnosti $\omega_x\, dV$ je hustota
→ diferenciál Lebesgueovy míry

^3

Parciální derivace na $\mathbb{R}^n$

- prostor bodů A
- prostor směrů $\Delta\vec{x}$
- f funkce na bodech $f(A)$

⇒ změna f při posunu

$$f(x_0 + \varepsilon \Delta\vec{x}) = f(x_0) + \varepsilon\, df(x_0) + O(\varepsilon^2)$$

$$df(x_0) = \frac{d}{d\varepsilon} f(x_0 + \varepsilon\Delta\vec{x})\Big|_{\varepsilon=0} = \sum_a \underbrace{\frac{\partial f}{\partial x^a}(x_0)}_{\text{parciální derivace}} \Delta x^a$$

Variace = ∂ v nekonečné dimenzi

- prostor historií /funkcí ... H, $\varphi, \psi, \dots$
- směry na H ... $\delta\varphi, \delta\psi$
- funkcionály na H ... $F[\varphi]$

⇒ změna F při posunu

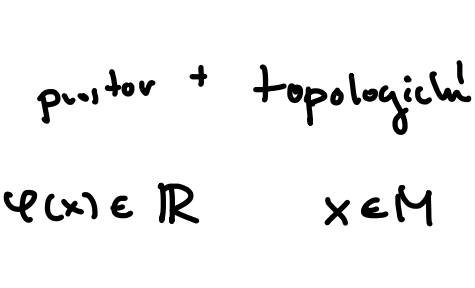
$$F[\varphi_0 + \varepsilon\delta\varphi] = F[\varphi_0] + \varepsilon\,\delta F[\varphi_0] + O(\varepsilon^2)$$

$$\delta F[\varphi_0] = \frac{d}{d\varepsilon} F[\varphi_0 + \varepsilon\delta\varphi_0]\Big|_{\varepsilon=0} = \int_x \underbrace{\frac{\delta F}{\delta\varphi^x}(\varphi_0)}_{\text{variace}} \delta\varphi^x\, dV$$

Skalární pole

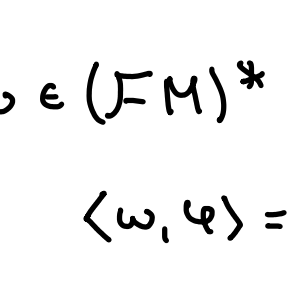
$$\underset{\text{prostoročas}}{\psi: M} \to \mathbb{R} \qquad g = -dt^2 + dx^2 + dy^2 + dz^2$$
$$\Omega = \langle \Sigma_1 | \Sigma_2 \rangle$$

akce
$$S_\Omega[\psi] = -\int_\Omega \left(\tfrac{1}{2}\nabla\psi^2 + V(\psi)\right) d\Omega$$



∇_μ prostoročasový gradient

$$V(\psi) = \tfrac{1}{2}m^2\psi^2 + \tfrac{1}{4!}\lambda\psi^4 \quad ?$$



} není vidět, co se tu děje

$$S[\psi_0 + \varepsilon\psi] = -\int_\Omega \left(\tfrac{1}{2}(\nabla\psi_0)^2 + V(\psi_0)\right) d\Omega$$
$$- \varepsilon \int_\Omega \left(\nabla\psi_0 \cdot \nabla\delta\psi + V'(\psi_0)\,\delta\psi\right) d\Omega + O(\varepsilon^2)$$

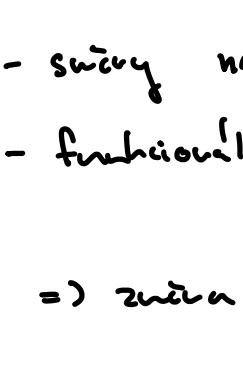
$$\delta S[\psi_0] = -\int_{\Omega_0} \left(\nabla\psi_0 \cdot \nabla\delta\psi + V'(\psi_0)\,\delta\psi\right) d^4\Omega$$
$$= -\int_\Omega \left(\nabla\cdot(\nabla\psi_0\,\delta\psi) - (\nabla\cdot\nabla\psi_0)\,\delta\psi + V'(\psi_0)\,\delta\psi\right) d^4\Omega$$
$$= -\underbrace{\int_{\partial\Omega} \nabla\psi_0\,\delta\psi\, d^3\Sigma}_{=0} - \int_\Omega \left(\Box\psi_0 + V'(\psi_0)\right)\delta\psi\, d^4\Omega$$

$\delta\psi|_\Sigma = 0$

$\Box = -\nabla_\mu\nabla^\mu$
$= -g^{\mu\nu}\nabla_\mu\nabla_\nu$

$$\Rightarrow \quad \frac{\delta S_\Omega}{\delta\psi}[\psi_0] = -\Box\psi_0 - V'(\psi_0) = 0$$

pro $V = \tfrac{1}{2}m^2\psi^2$ ⇒ Klein-Gordonova rovnice



$$\psi(x) = \varphi(t, x) = \varphi^x(t)$$
$$x = [t, x]$$

$$\nabla\psi = dt\,\frac{\partial\psi}{\partial t} + \vec{\nabla}\psi$$
$$= dt\,\dot{\varphi} + \vec{\nabla}\varphi$$

$$(\nabla\psi)^2 = -\dot{\psi}^2 + (\vec{\nabla}\psi)^2$$

$$S[\psi(t)] = \int_{t_2}^{t_1}\int_{\Sigma_t} \left(\tfrac{1}{2}\dot{\psi}^2 - \tfrac{1}{2}(\vec{\nabla}\psi)^2 - V(\psi)\right) \underbrace{dV\, dt}_{d\Omega}$$

$$S[\psi(t)] = \int_{t_2}^{t_1} L(\psi(t), \dot{\psi}(t))\, dt$$

$$L(\psi, \underset{\dot{\psi}}{\theta}) = \int_{\Sigma_t} \Big(\underbrace{\tfrac{1}{2}\theta^2}_{\text{kinetický člen}} - \underbrace{\tfrac{1}{2}\vec{\nabla}\psi^2 - V(\psi)}_{\text{potenciální člen}}\Big)\, dV$$

$$\Rightarrow \quad \frac{d}{dt}\left(\frac{\delta L}{\delta\theta}(\psi, \dot{\psi})\right) - \frac{\delta L}{\delta\psi}(\psi, \dot{\psi}) = 0$$

$\theta \to \theta + \varepsilon\delta\theta \qquad \psi \to \psi$

$$\delta L = \int_{\Sigma_t} \theta\,\delta\theta\, dV \quad \Rightarrow \quad \frac{\delta L}{\delta\theta}(\psi, \theta) = \theta$$

$\theta \to \theta \qquad \psi \to \psi + \varepsilon\delta\psi$

$$\delta L = \int_{\Sigma_t} \left(-\vec{\nabla}\psi\cdot\vec{\nabla}\delta\psi - V'(\psi)\,\delta\psi\right) dV$$
$$\overset{PP}{=} \int_{\Sigma_t} \left(\vec{\nabla}\cdot\vec{\nabla}\psi - V'(\psi)\right)\delta\psi\, dV$$

$$\Rightarrow \quad \frac{\delta L}{\delta\psi}(\psi, \theta) = \Delta\psi - V'(\psi)$$

$$\Rightarrow \quad \frac{d}{dt}(\dot{\psi}) - \Delta\psi + V'(\psi) = 0$$
$$\ddot{\psi} - \Delta\psi + V'(\psi) = 0$$
$$\Box\psi + V'(\psi) = 0$$

$$\pi(\psi, \theta) = \frac{\delta L}{\delta\theta}(\psi, \theta) = \theta$$
$$\theta = \pi$$

$$H(\psi, \pi) = \int_{\Sigma_t} \theta(\psi, \pi)\,\pi\, dV - L(\psi, \theta(\psi, \pi))$$
$$= \int_\Sigma \left(\pi^2 - \tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\vec{\nabla}\psi)^2 + V(\psi)\right) dV$$
$$= \int_\Sigma \left(\tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\vec{\nabla}\psi)^2 + V(\psi)\right) dV$$

$$\dot{\psi} = \frac{\delta H}{\delta\pi} = \pi \qquad \dot{\pi} = -\frac{\delta H}{\delta\psi} = \Delta\psi - V'(\psi)$$

$$\delta H = \int \left(\nabla\psi\cdot\nabla\delta\psi + V'(\psi)\,\delta\psi\right) dV$$
$$= \int \left(-\Delta\psi + V'(\psi)\right)\delta\psi\, dV$$

$$\to \quad \ddot{\psi} - \Delta\psi + V'(\psi) = 0$$
$$\Rightarrow \quad \Box\psi + V'(\psi) = 0$$

$$\{\psi^x, \pi_y\} = \delta^x_y = \delta(x|y)$$

$$\{F, G\} = \int_\Sigma \left(\frac{\delta F}{\delta\psi}\frac{\delta G}{\delta\pi} - \frac{\delta F}{\delta\pi}\frac{\delta G}{\delta\psi}\right) dV$$

funkce od x

Akce EM pole

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$$
$$S_{EM}[A] = -\frac{\varepsilon_0}{4}\int_\Omega F_{\alpha\beta}F^{\alpha\beta}\, d^4\Omega$$

zadaný externí zdroj

lineární interakce s potenciálem

$S_{ext}[A] \qquad$ Př.: $\quad S_{ext}[A] = \int_\Omega J^\mu_{ext} A_\mu\, d^4\Omega$

$$\frac{\delta S}{\delta A_\alpha} = -\varepsilon_0\nabla_\mu F^{\alpha\mu} + \frac{\delta S_{ext}}{\delta A_\alpha}$$
$$= -\varepsilon_0\nabla_\mu F^{\alpha\mu} + J^\alpha_{ext} \overset{?}{=} 0$$

$$\Rightarrow \quad \nabla_\mu F^{\alpha\mu} = \frac{1}{\varepsilon_0} J^\alpha_{ext}$$
$$+ \quad \nabla_{[\gamma} F_{\alpha\beta]} = 0$$

Lagrangeovský formalismus

$$A^\alpha = \begin{bmatrix} \phi \\ \vec{A} \end{bmatrix} \qquad F_{\mu\nu} = \begin{bmatrix} 0 & -\vec{E} \\ \vec{E} & \overleftrightarrow{B} \end{bmatrix}$$

$$\vec{E} = -\vec{\nabla}\phi - \dot{\vec{A}} \qquad \vec{B} = \vec{\nabla}\times\vec{A}$$

$$-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}(E^2 - B^2)$$

$$S_{EM}[\phi, \vec{A}] = \int_{t_2}^{t_1}\int_{\Sigma_t} \frac{\varepsilon_0}{2}(E^2 - B^2)\, dV dt$$

externí pole musí splňovat rovnici kontinuity

$$0 = \nabla_\alpha J^\alpha \qquad J = \begin{bmatrix} \varrho \\ \vec{j} \end{bmatrix}$$
$$\Rightarrow \quad \dot{\varrho} + \vec{\nabla}\cdot\vec{j} = 0$$

$$L_{EM}(\phi, \vec{A}, \dot{\phi}, \dot{\vec{A}}) = \frac{\varepsilon_0}{2}\int_{\Sigma_t} (E^2 - B^2)\, dV$$

$$L_{ext}(\phi, \vec{A}) \qquad L_{ext} = \int_{\Sigma_t} \left(-\varrho_{ext}\phi + \vec{j}_{ext}\cdot\vec{A}\right) dV$$

$$\frac{d}{dt}\left(\frac{\delta L}{\delta\dot{\vec{A}}}\right) - \frac{\delta L}{\delta\vec{A}} = 0 \quad \wedge \quad \frac{d}{dt}\left(\frac{\delta L}{\delta\dot{\phi}}\right) - \frac{\delta L}{\delta\phi} = 0$$

$$\vec{j} = -\cancel{\frac{\delta L_{ext}}{\delta\dot{\vec{A}}}} + \frac{\delta L_{ext}}{\delta\vec{A}} \qquad \varrho = \cancel{\frac{\delta L_{ext}}{\delta\dot{\phi}}} - \frac{\delta L_{ext}}{\delta\phi}$$

$$\left(\frac{\delta L_{EM}}{\delta\dot{\vec{A}}}\right)^{\boldsymbol{\cdot}} - \frac{\delta L_{EM}}{\delta\vec{A}} = \vec{j} \qquad \left(\frac{\delta L_{EM}}{\delta\dot{\phi}}\right)^{\boldsymbol{\cdot}} - \frac{\delta L_{EM}}{\delta\phi} = -\varrho$$

$$\dot{\vec{A}} \to \dot{\vec{A}} + \delta\dot{\vec{A}} \qquad \delta L_{EM} = \varepsilon_0\int \vec{E}\cdot(-\delta\dot{\vec{A}})\, dV$$

$$\Rightarrow \quad \frac{\delta L_{EM}}{\delta\dot{\vec{A}}} = -\varepsilon_0\vec{E}$$

$$\vec{A} \to \vec{A} + \delta\vec{A} \qquad \delta L_{EM} = -\varepsilon_0\int_\Sigma \vec{B}\cdot(\vec{\nabla}\times\delta\vec{A})\, dV$$
$$= \varepsilon_0\underbrace{\int_\Sigma \vec{\nabla}\cdot(\vec{B}\times\delta\vec{A})\, dV}_{\substack{\text{na okraj} \\ \Rightarrow 0}} - \varepsilon_0\int_\Sigma (\vec{\nabla}\times\vec{B})\cdot\delta\vec{A}\, dV$$

$$\Rightarrow \quad \frac{\delta L_{EM}}{\delta\vec{A}} = -\varepsilon_0\vec{\nabla}\times\vec{B}$$

$$\delta\dot{\phi} = 0 \quad \Rightarrow \quad \frac{\delta L_{EM}}{\delta\dot{\phi}} = 0$$

$$\delta\phi \qquad \delta L_{EM} = \varepsilon_0\int_\Sigma \vec{E}\cdot(-\vec{\nabla}\delta\phi)\, dV$$
$$= \varepsilon_0\int_\Sigma (\vec{\nabla}\cdot\vec{E})\,\delta\phi\, dV$$

$$\Rightarrow \quad \frac{\delta L_{EM}}{\delta\phi} = \varepsilon_0\vec{\nabla}\cdot\vec{E}$$

$$\Rightarrow \quad -\dot{\vec{E}} + \vec{\nabla}\times\vec{B} = \frac{1}{\varepsilon_0}\vec{j} \qquad \vec{\nabla}\cdot\vec{E} = \frac{1}{\varepsilon_0}\varrho$$
$$\vec{\nabla}\cdot\vec{B} = 0 \qquad \vec{\nabla}\times\vec{E} = -\dot{\vec{B}}$$

Hamiltonovský formalismus

$$L(\phi, \vec{A}, \dot{\phi}, \dot{\vec{A}}) = \frac{\varepsilon_0}{2}\int_\Sigma (E^2 + B^2)\, dV$$
$$+ \int_\Sigma \left(-\varrho_{ext}\phi + \vec{j}_{ext}\cdot\vec{A}\right) dV$$

$$\vec{E} = -\dot{\vec{A}} - \vec{\nabla}\phi \qquad \vec{B} = \vec{\nabla}\times\vec{A}$$

$$\vec{\pi} = \frac{\delta L}{\delta\dot{\vec{A}}} = -\varepsilon_0\vec{E} = \varepsilon_0(\dot{\vec{A}} + \vec{\nabla}\phi)$$

$$\kappa = \frac{\delta L}{\delta\dot{\phi}} = 0 \qquad \text{vazba – primární}$$

$\kappa(x) = 0 \qquad \forall x \in \Sigma$
$\kappa_\nu = 0 \qquad \forall \nu \in \mathcal{F}\Sigma \qquad \kappa_\nu = \int_\Sigma \kappa\nu\, dV$

zhlazení!

Fázový prostor

$\phi, \vec{A}, \kappa, \vec{\pi}$

$$\{F_1, F_2\} = \int \left(\frac{\delta F_1}{\delta\phi}\frac{\delta F_2}{\delta\kappa} - \frac{\delta F_1}{\delta\kappa}\frac{\delta F_2}{\delta\phi} + \frac{\delta F_1}{\delta\vec{A}}\cdot\frac{\delta F_2}{\delta\vec{\pi}} - \frac{\delta F_1}{\delta\vec{\pi}}\cdot\frac{\delta F_2}{\delta\vec{A}}\right) dV$$

z nějakého důvodu je zajímavé:

$\{\kappa_{\nu_1}, \kappa_{\nu_2}\} = 0$
$\{\kappa(x), \kappa(y)\} = 0 \qquad x, y \in \Sigma$ } vazby první třídy

Jsou důsledkem odlišné od vazby druhé třídy
→ zde se vyskytují jiné když se symetrizuje

$$\dot{\vec{A}} = \frac{1}{\varepsilon_0}\vec{\pi} + \vec{\nabla}\phi$$

$\dot{\phi} = \lambda$ → volím si libovolně, protože se tam nevyskytuje

⇒ něco jako Lagrangeův multiplikátor

$$H_\lambda = \int_\Sigma \left(\dot{\phi}\kappa + \dot{\vec{A}}\cdot\vec{\pi}\right) dV - L(\phi, \dot{\phi}, \vec{A}, \dot{\vec{A}})$$
$$= \int_\Sigma \Big(\lambda\kappa + \left(\tfrac{1}{\varepsilon_0}\vec{\pi} - \vec{\nabla}\phi\right)\cdot\vec{\pi} - \frac{1}{2\varepsilon_0}\vec{\pi}^2$$
$$+ \frac{\varepsilon_0}{2}\vec{B}^2 + \varrho_{ext}\phi - \vec{j}_{ext}\cdot\vec{A}\Big)\, dV$$

$$= \int_\Sigma \left(\frac{1}{2\varepsilon_0}\vec{\pi}^2 + \frac{\varepsilon_0}{2}\vec{B} - \vec{j}_{ext}\cdot\vec{A} + \phi(\vec{\nabla}\cdot\vec{\pi} + \varrho_{ext}) + \lambda\kappa\right) dV$$

$$\frac{\delta H_\lambda}{\delta\phi} = \vec{\nabla}\cdot\vec{\pi} + \varrho_{ext} \qquad \frac{\delta H_\lambda}{\delta\vec{A}} = \varepsilon_0(\vec{\nabla}\times\vec{B}) - \vec{j}_{ext} = \dot{\vec{\pi}}$$

$$\frac{\delta H}{\delta\kappa} = \lambda = \dot{\phi} \qquad \frac{\delta H_\lambda}{\delta\vec{\pi}} = \frac{1}{\varepsilon_0}\vec{\pi} - \vec{\nabla}\phi = \dot{\vec{A}}$$

Pozorovatelné $\quad F[\phi, \vec{A}, \kappa, \vec{\pi}; t]$

$$\dot{F} = \{F, H_\lambda\} + \frac{\partial F}{\partial t}$$

$$\dot{\kappa}_\nu = \{\kappa_\nu, H_\lambda\} + \frac{\partial\kappa_\nu}{\partial t} \qquad \kappa_\nu = \int \nu\kappa\, dV \qquad \nu = \nu(t)$$

$$= -\int_\Sigma \underbrace{\frac{\delta\kappa_\nu}{\delta\kappa}}_{\nu}\frac{\delta H_\lambda}{\delta\phi}\, dV + \kappa_{\dot{\nu}}$$

$$= -\int_\Sigma \nu(\vec{\nabla}\cdot\vec{\pi} + \varrho_{ext})\, dV + \kappa_{\dot{\nu}} = 0$$

zachovat vazby

$$\Rightarrow \quad G = \vec{\nabla}\cdot\vec{\pi} + \varrho_{ext} \equiv 0$$
$$G_\nu = \int_\Sigma \nu G\, dV \qquad \text{Gaussova vazba}$$

[illegible]

$$\Rightarrow \quad \dot{\kappa}_\nu = G_\nu + \kappa_{\dot{\nu}}$$

$$\dot{G}_\nu = \{G_\nu, H_\lambda\} + \frac{\partial G_\nu}{\partial t}$$
$$= -\int \frac{\delta G_\nu}{\delta\vec{\pi}}\cdot\frac{\delta H_\lambda}{\delta\vec{A}}\, dV + \int_\Sigma \left(\nu\dot{\varrho}_{ext} + \dot{\nu}(\vec{\nabla}\cdot\vec{\pi} + \varrho_{ext})\right) dV$$
$$= \quad \vdots$$
$$= G_{\dot{\nu}} \qquad \text{vymizí pro } G = 0$$

algebra vazeb:

$$\{\kappa_\mu, \kappa_\nu\} = 0 \qquad \{G_\mu, G_\nu\} = 0 \qquad \{\kappa_\mu, G_\nu\} = 0$$

$$\dot{\phi} = \frac{\delta H_\lambda}{\delta\kappa} = \lambda \qquad \dot{\phi} = \lambda$$

$$\dot{\kappa} = -\frac{\delta H_\lambda}{\delta\phi} = G \qquad \kappa = 0 \qquad \vec{\nabla}\cdot\vec{E} = \frac{1}{\varepsilon_0}\varrho_{ext}$$

$$\dot{\vec{A}} = \frac{\delta H_\lambda}{\delta\vec{\pi}} = \frac{1}{\varepsilon_0}\vec{\pi} - \vec{\nabla}\phi \qquad \vec{E} = -\dot{\vec{A}} - \vec{\nabla}\phi$$

$$\dot{\vec{\pi}} = -\frac{\delta H_\lambda}{\delta\vec{A}} = -\varepsilon_0(\vec{\nabla}\times\vec{B}) + \vec{j}_{ext} \qquad \vec{\nabla}\times\vec{B} - \dot{\vec{E}} = \frac{1}{\varepsilon_0}\vec{j}_{ext}$$
$$\vec{\nabla}\cdot\vec{B} = 0$$

$$H_\lambda = \int_\Sigma \left(\frac{1}{2\varepsilon_0}\vec{\pi}^2 + \frac{\varepsilon_0}{2}\vec{B}^2 - \vec{j}_{ext}\cdot\vec{B}\right) + G_\phi + \kappa_\lambda$$

částečná redukce fázového prostoru

$$\phi = \int \lambda\, dt$$
$$\Rightarrow \quad \vec{A}, \vec{\pi}$$